Case Study on Preparation of a Software Project Plan for a Hotel Room Booking Application

Aim

To prepare a software project plan covering scope, timeline, and team roles for the development of a sample application, using a Hotel Room Booking Application as the reference case.

Objective

- Define the project scope, including included and excluded features.
- Break down the project into phases with a realistic timeline.
- Assign clear team roles and responsibilities.
- Identify key risks and plan mitigation strategies.
- Establish milestones to track project progress.

Selected Software Application

Hotel Room Booking Application

Overview

A Hotel Room Booking Application allows guests to search hotels, check room availability, book rooms, make payments, and manage reservations, while hotel staff manage inventory, pricing, and bookings from an admin panel.

Project Scope

In Scope

- Guest registration and login
- Hotel and room search with filters (location, price, rating)
- Real-time room availability and booking
- Online payment and invoice generation
- Booking cancellation and refund handling
- Hotel admin panel for inventory and pricing management
- Guest reviews and ratings

Out of Scope

- Third-party travel package bundling (flights, cabs)
- Loyalty/rewards program
- Multi-language support

Functional Requirements

Guest Module

- Register/Login
- Search Hotels and Rooms
- View Room Details and Photos
- Book Room and Make Payment
- Cancel/Modify Booking

Hotel Admin Module

- Manage Room Inventory and Pricing
- View and Manage Bookings
- Update Hotel Information

System Admin Module

- Manage Registered Hotels
- Monitor Transactions
- Generate Reports

Project Timeline

The project is planned across six phases over an estimated 16-week schedule.

Phase	Duration	Key Deliverables
Requirement Analysis	Week 1 - 2	Software Requirement Specification (SRS)
System Design	Week 3 - 4	Architecture diagram, database schema, UI wireframes
Development	Week 5 - 10	Guest module, admin module, payment integration
Testing	Week 11 - 13	Test cases, bug reports, UAT sign-off
Deployment	Week 14	Production release on cloud hosting
Maintenance	Week 15 - 16 onward	Bug fixes, monitoring, minor enhancements

Team Roles and Responsibilities

Role	Responsibility
Project Manager	Plans timeline, tracks progress, manages risks and stakeholder communication
Business Analyst	Gathers requirements and prepares the SRS document

Role	Responsibility
UI/UX Designer	Designs wireframes, prototypes, and the visual interface
Backend Developer	Builds APIs, booking logic, and database integration
Frontend Developer	Implements the guest-facing and admin-facing interfaces
QA Engineer	Designs and executes test cases, logs and verifies bug fixes
DevOps Engineer	Sets up CI/CD pipeline, deployment, and monitoring

Risk Management

Risk	Impact	Mitigation
Delay in payment gateway integration	Medium	Start integration early with a sandbox environment
Inaccurate room availability data	High	Implement real-time locking to prevent double booking
Scope creep from stakeholders	Medium	Freeze scope after sign-off; log new requests for later phases
Server downtime during peak booking season	High	Use auto-scaling cloud infrastructure and load testing

Milestones

- M1: SRS and design sign-off — Week 4
- M2: Core booking flow functional (dev complete) — Week 10
- M3: Testing and UAT complete — Week 13
- M4: Go-live — Week 14

Results

- A clearly defined scope prevented ambiguity between guest-facing and admin-facing features.
- A phase-wise timeline gave the team measurable checkpoints for tracking progress.
- Clear role assignment reduced overlap and improved accountability across the team.
- Early risk identification, especially around payments and room-inventory accuracy, minimized delays.

Conclusion

A well-structured project plan is essential for the successful delivery of a software application. By clearly defining the scope, setting a realistic timeline, and assigning well-defined team roles, the development of the Hotel Room Booking Application can proceed with better coordination, fewer delays, and higher overall quality.